

BT101

Introductory Biology

(Lecture 6)

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Introduction



Evolution of life:

- ❑ Origin of Life;
- ❑ Darwin's concepts of evolution
- ❑ **Biodiversity**

Cell, the structural and functional unit of life:

- ❑ Three domains of life
- ❑ Cell types, cell organelles and structure
- ❑ Basic biomolecules of cell

What is Biodiversity?



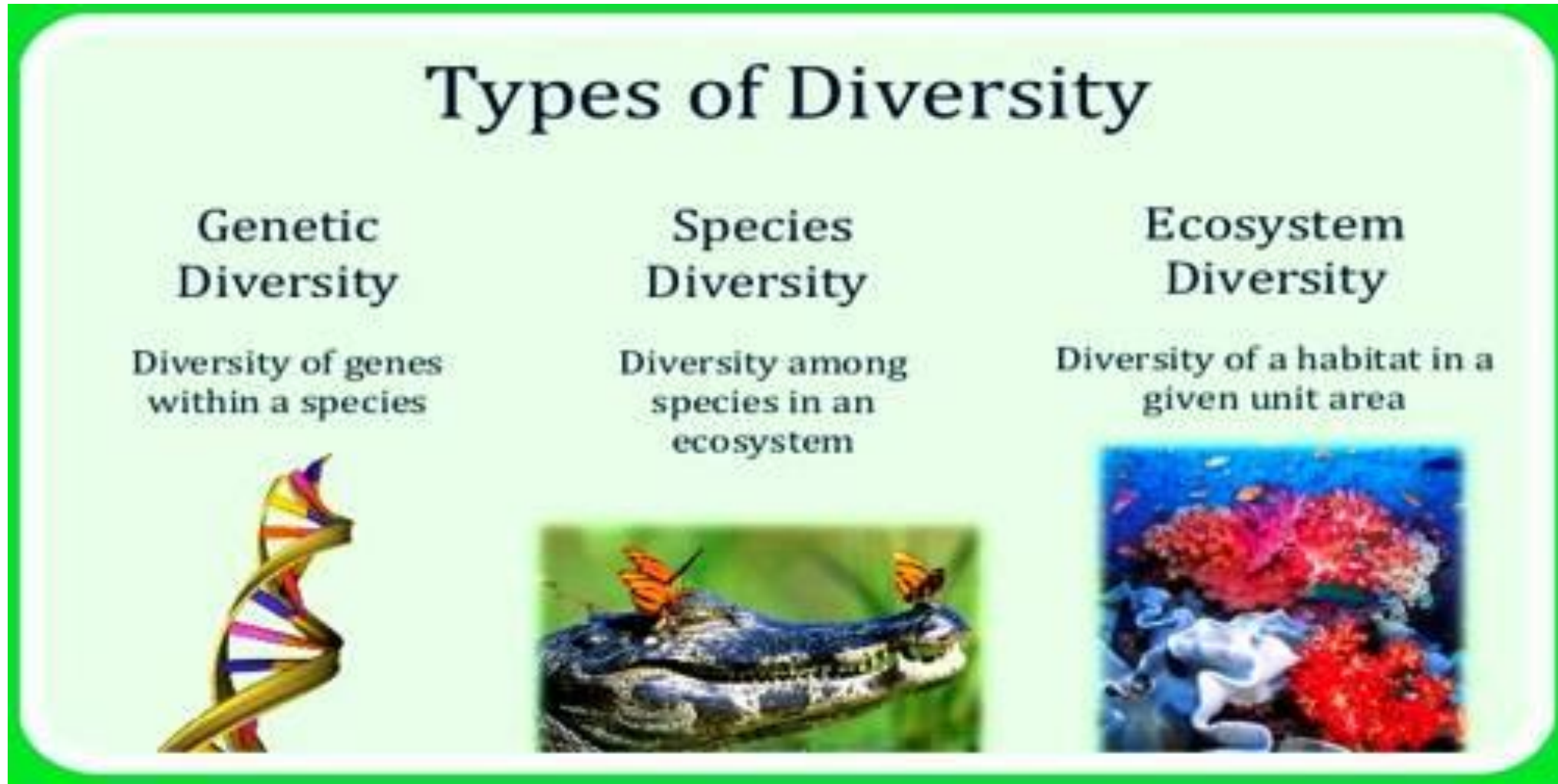
- ❑ The term Biodiversity was first coined by **Walter G. Rosen in 1985**.
- ❑ Biodiversity, is a short term for biological diversity, refers to all of the population, species and communities in a defined area.
- ❑ Biodiversity is the variation among living organisms from different sources including terrestrial, marine and desert ecosystems, and the ecological complexes of which they are a part.
- ❑ Biodiversity forms the foundation of the vast array of ecosystem services, that critically contribute to human well-being.
- ❑ It typically measures the variation at genetic, species, and ecosystem level.

Types of Biodiversity

1. Genetic diversity

2. Species diversity

3. Ecosystem diversity



1. Genetic Diversity

- Genetic diversity is the **variation in the genetic composition** of individuals in a population, community or species.
- **Allows individuals to adapt to different conditions. Thus, high genetic diversity increases the ability of the populations and species to survive in the major changes in their environment such as the Climate Change.**



2. Species Diversity

- Species diversity is the variety of species in a particular habitat or ecosystem.
- The richness of species in an ecosystem is usually referred to as Species Diversity.



The diversity of the smaller organisms (e.g. Phytoplankton, the microscopic marine algae) is less known than the larger organisms (e.g. Mammals such as Dolphins and Whales).

3. Ecosystem Diversity



- Ecosystem diversity describes the variation in all **Living and Non-living** things in a **particular Geographic or Ecological Region**.
- Ecosystems comprise unique combinations of animals, plants, micro-organisms and physical characteristics that define a location.
- For example, a small pond constitute an ecosystem and possesses a set of flora and fauna different from a river which is another type of ecosystem.

Ecosystem Diversity (Terrestrial and Aquatic)

Grass land ecosystem



Forest Ecosystem



Desert Ecosystem



Marine Ecosystem



Pond Ecosystem



River Ecosystem



Why is Biodiversity important?



- ☐ Oxygen to breathe
- ☐ Food to eat
- ☐ Clothes to wear
- ☐ House to stay

Why is Biodiversity important?



- ❑ Everything that exists in an ecosystem is part of the web of life, including humans.
- ❑ Plant, animal, and insect species interact and depend upon one another for what each offers, such as food, shelter, oxygen, and soil enrichment.
- ❑ "It is reckless (hasty) to suppose that biodiversity can be diminished indefinitely without threatening humanity itself." - **Edward O. Wilson** (Father of Biodiversity)
- ❑ At the ecosystem level, biodiversity provides the conditions and drives the processes that sustain the global economy- and our survival as a species.

Magnitude of Biodiversity

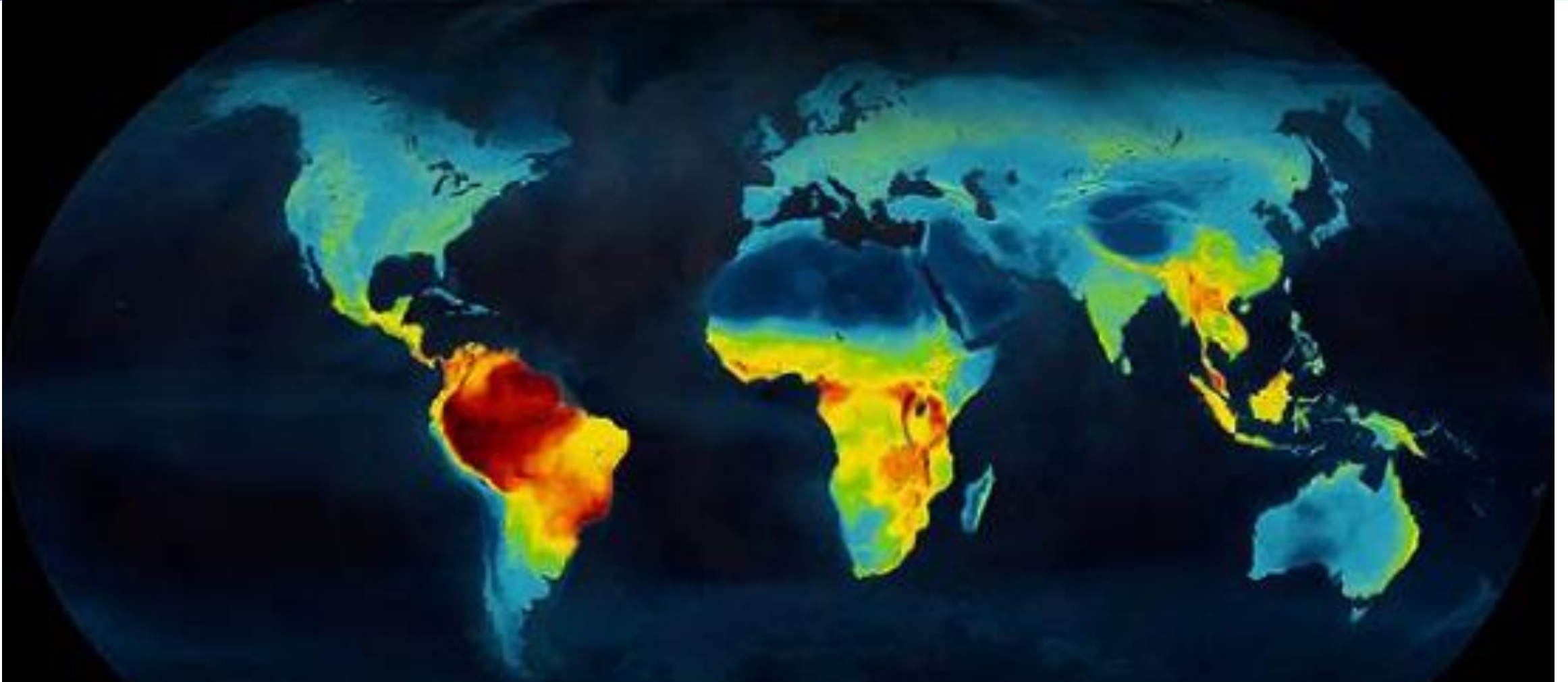


- ❑ The known and described number of species on earth is between 1.7 and 1.8 million.
- ❑ **Current estimates of the total number of species range from 5 to 30 million.**
- ❑ About 61% of these species are insects.

Table lists the Approximate number of species on earth.

GROUP	Number of Species
MAMMALS (Human, Bat, Whale etc.)	4650
BIRDS	9700
REPTILES (crawls, moves on its belly or on short legs. e.g. Snake, Corcodile, lizard, etc.)	7150
FISH	27000
AMPHIBIANS (Can live both on land and in water, Frog, etc.)	4780
INSECTS	1025000
HIGHER PLANTS	270000
ALGAE	40000
PROTOZOANS (Single celled organisms e.g. <i>Amoeba</i> , <i>Paramecium</i>)	40000

Magnitude of Biodiversity



Distribution of **living terrestrial vertebrate** species, highest concentration of diversity shown in red in equatorial regions, declining polewards (towards the blue end of the spectrum) (Mannion 2014)



Benefits of Biodiversity



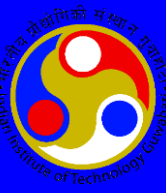
❑ **Consumptive value:**

- Food/Drink
- Fuel
- Medicine: painkillers, antibiotics, anticancer drugs
- Crop varieties
- Industrial Material: paper, rubber

❑ **Non-Consumptive Value:**

- Recreation: visiting beach, whale watching, photography, surfing, scuba diving and boating, where resource is not removed from Ocean or Coast.
- Education and Research
- Traditional value

Threats to Biodiversity



Natural causes:

- Narrow Geographical Area
- Over Population
- Low Breeding rate
- Natural Disasters

Anthropogenic (by Human) causes:

- Pollution
- Hunting
- Global warming and climate change
- Agriculture

Threats to Biodiversity



Biodiversity is under serious threat as a result of **Human activities**.

The main dangers worldwide are

- i) **population growth and resource consumption**
- ii) **climate change and global warming**
- iii) **habitat conversion and urbanization**
- iv) **overexploitation of natural resources and**
- v) **environmental degradation**
- vi) **etc.**

Why is Biodiversity important?



Important Links to see Videos



- ❑ **Biodiversity:**
- ❑ https://www.youtube.com/watch?v=GK_vRtHJZu4&t=2s
- ❑ https://www.youtube.com/watch?v=ewJImPSV_XE (in hindi)